# THE SPECTRAL SEQUENCE OF A SIMPLICIAL SITE FOR SHEAF COHOMOLOGY

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: in-source stubs lemma-cech-complex, lemma-restriction-to-components-site, lemma-restriction-injective-to-component-site; external tools from Homology (first-quadrant spectral sequence, double-complex resolution).

*Editorial note.* The label lemma-simplicial-sheaf-cohomology-site names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- Cech complex of the simplicial site. Same unit, Lemma lemma-cech-complex. Provides the horizontal arrows of the double complex (the simplicial cohomology complex).
- Exactness of restriction. Same unit, Lemma lemma-restriction-to-components-site. Restriction to C_p is exact, so I_p^bullet represents K_p.
- Injectivity of restrictions. Same unit, Lemma lemma-restriction-injective-to-component-site. The sheaves I_p^q are injective abelian sheaves on C_p, so column cohomology computes H^q(C_p, K_p).

*Source reference.* Original lemma 0133: sources/stacks-chow/tags.txt, source lines 1421–1421. Prerequisite pointer invoked at proof line(s) 1809; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0132: sources/stacks-chow/tags.txt, source lines 1420–1420. Prerequisite pointer invoked at proof line(s) 1808; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 03OZ: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1752–1752. Prerequisite pointer invoked at proof line(s) 1795; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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*Date*: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original reference 09WH: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1086–1086. Prerequisite pointer invoked at proof line(s) 1805; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 09WG: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 970–970. Prerequisite pointer invoked at proof line(s) 1801; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 09WE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 888–888. Prerequisite pointer invoked at proof line(s) 1778; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *In Situation 09WE. For $K$ in $D^+(\mathcal{C}_{total})$ there is a spectral sequence $(E_r, d_r)_{r\geq 0}$ with*

$$E_1^{p,q} = H^q(\mathcal{C}_p, K_p), \quad d_1^{p,q} : E_1^{p,q} \to E_1^{p+1,q}$$

*converging to $H^{p+q}(\mathcal{C}_{total}, K)$. This spectral sequence is functorial in $K$.*

*Proof.* Let $\mathcal{I}^\bullet$ be a bounded below complex of injectives representing $K$. Consider the double complex with terms

$$A^{p,q} = \Gamma(\mathcal{C}_p, \mathcal{I}_p^q)$$

where the horizontal arrows come from Lemma 03OZ and the vertical arrows from the differentials of the complex $\mathcal{I}^\bullet$. The rows of the double complex are exact in positive degrees and evaluate to $\Gamma(\mathcal{C}_{total}, \mathcal{I}^q)$ in degree 0. On the other hand, since restriction to $\mathcal{C}_p$ is exact (Lemma 09WG) the complex $\mathcal{I}_p^\bullet$ represents $K_p$ in $D(\mathcal{C}_p)$. The sheaves $\mathcal{I}_p^q$ are injective abelian sheaves on $\mathcal{C}_p$ (Lemma 09WH). Hence the cohomology of the columns computes the groups $H^q(\mathcal{C}_p, K_p)$. We conclude by applying Homology, Lemmas 0132 and 0133. $\square$